MILITARY SPECIFICATION

HOSE AND HOSE ASSEMBLIES, CHEMICAL (INDUSTRIAL AND FIRE FIGHTING)

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. Scope. This specification covers hose and hose assemblies used for conveying liquids of a chemical nature and for use with chemical fire fighting equipment.

1.2 Classification. Hose and hose assemblies shall be of the following types, forms, classes, and sizes, as specified (see 6.2):

Type I - Chemical fire fighting.
Type II - Industrial chemical.
Form A - Braided.
Form B - Wrapped.
Class a - With couplings.
Class b - Without couplings.
Size 3/4 - 3/4-inch inside diameter (ID).
Size 1 - 1-inch ID.
Size 1-1/4 - 1-1/4 inch ID.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

WW-C-624	- Coupling Assembly, Hose (Garden, Water and Water Suction).

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MIL-P-775	- Packaging of Hose, Hose Assemblies, Rubber, Plastic, Fabric or Metal (Including Tubing); and Fittings, Nozzles, and Strainers.
MIL-C-11053	- Coupling Assemblies, Hose (Chemical, 3/4- and 1-Inch).

STANDARDS

FEDERAL

Fed. Test Method Std. No. 601	- Rubber, Sampling, and Testing.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASTM D 380	- Rubber Hose, Testing.
ANSI/ASTM D 412	- Rubber Properties in Tension.
ANSI/ASTM D 413	- Rubber Property-Adhesion to Flexible Substrate.
ANSI/ASTM D 572	- Rubber Deterioration by Heat and Oxygen Pressure.
ANSI/UL 92	- Fire Extinguisher and Booster Hose.

(Application for copies should be addressed to the American National Standards Institute, Inc. 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The hose and hose assemblies shall consist of a rubber tube, braided or wrapped fabric reinforcement, a rubber cover, and when applicable, a coupling on each end as specified herein.

3.2 First article (first produced hose and hose assemblies). The contractor shall furnish one or more hose and hose assemblies as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce hose and hose assemblies that will comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval of the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.4 Length. The length of the hose and hose assemblies shall be 50 feet or 100 feet as specified (see 6.2). Length tolerance shall be plus or minus 1.0 percent.

3.5 Hose assemblies. The hose assembly shall consist of a length of hose fitted with a coupling at each end. Couplings shall conform to WW-C-624 or MIL-C-11053, type and size as specified (see 6.2). Female couplings shall be furnished with washers.

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3.5.1 Coupling attachment. Couplings shall be attached to the hose in a manner that will not damage the hose lining or cover and when tested as specified in 4.5.2.2 shall withstand a hydrostatic pressure of not less than 400 pounds per square inch (psi) without separation from the hose, slippage, or leakage between the hose and coupling.

3.6 Marking. Each length of hose shall be indelibly marked with manufacturer's name, month and year of manufacture, and with the words "TESTED TO 400 PSI". Lengths of 50 feet shall have two markings per section approximately 25 feet apart. Lengths over 50 feet shall have the markings spaced not more than 27 feet apart.

3.7 Type I, chemical fire fighting hose.

3.7.1 Certification. The contractor shall submit to the contracting officer proof that the hose he proposes to supply under this specification conforms to the requirements of ANSI/UL 92. The label or listing of the Underwriters' Laboratories Inc., may be accepted as evidence that the hose conforms to this requirement. In lieu of the label or listing, the contractor may submit independent proof satisfactory to the contracting officer that the hose conforms to the applicable requirements of the published standards including methods of test of ANSI/UL 92.

3.8 Type II, Industrial chemical hose.

3.8.1 Tube and cover. The inner tube and outer cover shall be natural rubber, synthetic rubber compound, or a mixture thereof.

3.8.2 Reinforcements.

3.8.2.1 Form A, braided. Plies of reinforcement shall be cotton yarn, rayon, or other synthetic fibers. Braids shall be impregnated with rubber compound for joining the plies to the tube and cover. There shall be a distinct layer of rubber compound between the plies of reinforcement. The number of plies shall be in accordance with Table I.

3.8.2.2 Form B, wrapped. Plies of reinforcement shall be cotton duck, rayon, or other synthetic fibers. The wrapped fabric shall be evenly and firmly woven. The layers shall be applied with the edges lapped at least 1/2 inch and shall be impregnated with rubber compound for joining the plies to the tube and cover. The number of plies shall be in accordance with Table I.

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3.8.3 Physical requirements. Hose shall meet the requirements of Table I.

TABLE I. Physical requirements.

Size, inside diameter, inches	3/4	1	1-1/4
Tolerance, inside diameter, plus or minus, inch	1/32	1/16	1/16
Outside diameter, maximum, inches	1-5/16	1-5/8	1-7/8
Thickness, minimum, inch:			
Tube	0.060	0.060	0.060
Cover	0.047	0.047	0.047
Reinforcement, total plies:			
Braided	3	3	3
Wrapped	4	4	4
Proof pressure, psi, minimum	400	400	400
Friction, minimum pounds:			
Between cover and plies:	10	10	10
Between tube and plies	10	10	10
Between plies	12	12	12
Tensile strength before aging, minimum pounds per square inch:			
Tube	1,000	1,000	1,000
Cover	1,200	1,200	1,200
Ultimate elongation, before aging, minimum percent:			
Tube	250	250	250
Cover	300	300	300

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3.8.4 Accelerated aging. Tensile strength and ultimate elongation of the tube and cover after accelerated aging shall be not less than 60 percent of the tensile strength and elongation of the original values before aging.

3.9 Workmanship. Hose shall be uniform in thickness. The tube and cover shall be smooth, free from pits and blisters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced hose and hose assembly inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced hose and hose assembly inspection.

4.3.1 Examination. The first produced hose and hose assembly shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection of the applicable hose or hose assembly.

4.3.2 Tests. The first produced hose and hose assembly shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection of the applicable hose or hose assembly.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1 and 4.5.2.2. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The hose and hose assembly shall be examined as specified herein for the following defects:

Major

101. Type and class not as specified.
102. Yarn not evenly and firmly braided.
103. Fabric not evenly and firmly woven.
104. Fabric sewed instead of lapped.
105. Yarn or fabric not thoroughly saturated with rubber.
106. Incorrect size.
107. Tolerances not within specified limits.
108. Workmanship not as specified.

Minor

201. Marking missing, incomplete, or illegible.

4.5.2 Tests.

4.5.2.1 Length. Hose and hose assembly length shall be determined in accordance with Fed. Std. No. 601, Method 2411. Hose assembly length shall be the distance between the inside edges of the couplings. Nonconformance to 3.4 shall constitute failure of this test.

4.5.2.2 Coupling attachment. The hose assembly shall be tested in accordance with ANSI/ASTM D 380, Sections 12, 13, 15.1, and 15.3, except that the hydrostatic pressure shall be 400 psi and held for 2 minutes. Nonconformance to 3.5.1 shall constitute failure of this test.

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4.5.2.3 Type I hose. Chemical fire fighting hose shall be tested in accordance with ANSI/UL 92 (see 3.7).

4.5.2.4 Type II hose.

4.5.2.4.1 Diameter. The inside diameter tolerance and the maximum outside diameter of the hose shall be determined in accordance with ANSI/ASTM D 380. Nonconformance to Table I shall constitute failure of this test.

4.5.2.4.2 Thickness, tube and cover. The mimimum thickness of the tube and cover shall be determined in accordance with Fed. Std. No. 601, Method 2011. Nonconformance to Table I shall constitute failure of this test.

4.5.2.4.3 Proof pressure. The hose shall be tested in accordance with ANSI/ASTM D 380, except that the hydrostatic pressure shall be 400 psi and held for 2 minutes. Nonconformance to Table I shall constitute failure of this test.

4.5.2.4.4 Friction. Friction between cover and plies, tube and plies, and between plies shall be determined in accordance with ANSI/ASTM D 413, machine method. Nonconformance to Table I shall constitute failure of this test.

4.5.2.4.5 Tensile strength and elongation. The tensile strength and elongation characteristics shall be determined in accordance with ANSI/ASTM D 412. Nonconformance to Table I shall constitute failure of this test.

4.5.2.4.6 Accelerated aging. The deterioration of the tube and cover shall be determined in accordance with ANSI/ASTM D 572, with the time of aging 48 plus or minus 1/4 hours. Deterioration shall be measured by the change in tensile strength and elongation. Die F in ASTM D 412, shall be used for cutting the specimens. Nonconformance to 3.8.4 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking for Level A or Level B shall be examined to determine compliance with the Quality Assurance Provisions of MIL-P-775. The Commercial preservation, packing, and marking shall be examined to determine compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

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5.1.1 Level A. Hose and hose assemblies shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-P-775.

5.1.2 Commercial. Hose and hose assemblies shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Hose and hose assemblies, preserved as specified in 5.1, shall be packed in accordance with the Level A packing requirements of MIL-P-775.

5.2.2 Level B. Hose and hose assemblies, preserved as specified in 5.1, shall be packed in accordance with the Level B packing requirements of MIL-P-775.

5.2.3 Commercial. Hose and hose assemblies, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with the requirements specified in MIL-P-775.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Hose and hose assemblies are intended for conveying liquids of chemical nature, for use on fire extinguishers and fire fighting equipment.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Type, form, class, and size of hose required (see 1.2).
(c) Time frame required for the submission of the first produced hose and hose assembly and number of hose and hose assemblies required (see 3.2).
(d) Length of hose and hose assemblies (see 3.4).
(e) Type and size of couplings (see 3.5).
(f) Degree of preservation and degree of packing required (see 5.1 and 5.2).

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6.3 First produced hose and hose assemblies. Any changes or deviations of production hose or hose assemblies from the approved first produced hose and hose assembly during production will be subject to the approval of the contracting officer. Approval of the first produced hose and hose assembly will not relieve the contractor of his obligation to furnish hose and hose assemblies conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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